Problem Set for 03 April 2025

Definition (Eigenvector, Eigenvalue, and Eigenpair) Let $f : V \to V$ be an endomorphism. If and only if there exists

$$f(v) = \lambda v, \quad (\lambda \in \mathbb{F}, \quad v \neq 0),$$

we say:

- v is an eigenvector of f;
- λ is an eigenvalue of f; and
- (λ, v) is an eigenpair.

Exercise 1 Determine all eigenvalues of the linear endomorphism:

$$\varphi : \mathbb{R}[x] \to \mathbb{R}[x], \quad f(x) \mapsto \int_0^x f(t)\,\mathrm{d}\,t.$$

- Hint: the expression might be misleading, set $V := \mathbb{R}[x]$, one has

$$\varphi : V \to V, \quad f \mapsto \begin{bmatrix} \mathbb{R} & \to & \mathbb{R} \\ x & \mapsto & \int_0^x f(t)\,\mathrm{d}\,t \end{bmatrix}.$$

Exercise 2 Given $a, b \in \mathbb{R}$, determine all eigenpairs of the linear endomorphism:

$$\varphi : \mathbb{R}[x] \to \mathbb{R}[x], \quad f(x) \mapsto f(ax+b).$$

- Hint: find fixed point of $ax_0 + b = x_0$.

Exercise 3 Let V the space of smooth, bounded, and real valued functions over $\mathbb{R}$, which means that $f \in V$ maps $\mathbb{R}$ to some $[a, b] \subseteq \mathbb{R}$, and $\frac{\mathrm{d}^{\forall k \geq 1}}{\mathrm{d}\, x} f(x)$ exists. Determine all eigenpairs of the linear endomorphism:

$$\varphi_k : V \to V, \quad f(x) \mapsto \frac{\mathrm{d}^k}{\mathrm{d}\, x} f(x).$$

- Hint: for $P \in \mathbb{Z}[x]$ with distinct roots $\{z_k\}_{k=1}^{\deg n} \subseteq \mathbb{C}$, the solution of $P(\frac{\mathrm{d}}{\mathrm{d}\, x})(f) = 0$ forms a linear space spanned by $\{e^{z_i x}\}_{i=1}^{\deg n}$. For instance, $f'' - 3f' + 2f = 0$ has a linear space of the solutions $C_1 e^x + C_2 e^{2x}$ for $C_i \in \mathbb{R}$.
- **(Optional)** How would you like to define $\frac{\mathrm{d}^{1/2}}{\mathrm{d}\, x} f(x)$?

Exercise 4 Let $f : V \to V$ be an $\mathbb{F}$-linear endomorphism. Prove or disprove the following statement:

- v is an eigenvector of f, if and only if $v \in \ker(\lambda \cdot \mathrm{id} - f)$.

Exercise 5 (Optional) Let $V = C^0([0,1])$, i.e., the space of continuous real-valued functions over the closed interval $[0,1]$. Determine all eigenpairs of the operator:

$$V \to V, \quad f(x) \mapsto \int_0^1 K(x,y)\,\mathrm{d}\,y,$$

where the kernel function is defined as:

$$K(x,y) := \min(x,y) - x \cdot y.$$

- Hint: Under appropriate conditions,

$$\frac{\mathrm{d}}{\mathrm{d}\,x} \int_{f(x)}^{g(x)} H(x,y)\,\mathrm{d}\,y = H(x,g(x)) - H(x,f(x)) + \int_{f(x)}^{g(x)} H_x(x,y)\,\mathrm{d}\,y.$$

Exercise 6 (optional) Let V be the space of absolutely integrable functions from $\mathbb{R}$ to $\mathbb{C}$. Define the Fourier transform:

$$\mathcal{F} : V \to V, \quad f(x) \mapsto \widehat{f}(u) := \frac{1}{2\pi} \int_{-\infty}^{+\infty} f(t)\, \mathrm{e}^{-\mathrm{i}\, tu}\, \mathrm{d}\, t.$$

1. Is $\mathcal{F}$ a well-defined $\mathbb{C}$-linear endomorphism? If unsure, consult Professor Tao for appropriate spaces V, or simply take

$$W := \bigcap_{k \geq 0} V_k, \quad V_0 := V, \quad V_{k+1} := \mathcal{F}(V_k) \cap V_k$$

   which ensures that $\mathcal{F}(W) \subset W$ (show that $W \neq 0$?).
2. Determine all eigenvalues of $\mathcal{F}$ as an $\mathbb{R}$-linear transformation.
3. Determine all eigenvalues of $\mathcal{F}$ as a $\mathbb{C}$-linear transformation.